

Chapter 15: SWAP, Phase Shift, and CCNOT (Toffoli) Gates

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SWAP Gate

- A SWAP gate, as its name implies, swaps the numbers in the basis states of a 2-qubit register. If we consider only the basis states, it is equivalent to swapping the states of two electrons. The definition is

$$U_{SWAP}|ab\rangle = |ba\rangle$$

where a and b are just the numbers of the first and the second qubits in the basis states, respectively. They can be either 0 or 1. All the basis states can be applied:

$$U_{SWAP}|00\rangle = |00\rangle$$

$$U_{SWAP}|01\rangle = |10\rangle$$

$$U_{SWAP}|10\rangle = |01\rangle$$

$$U_{SWAP}|11\rangle = |11\rangle$$

Therefore, only $|01\rangle$ and $|10\rangle$ are changed (to each other) under the SWAP operation. For $|00\rangle$ and $|11\rangle$, they are unchanged because swapping “0” and “0” or “1” and “1” is equivalent to doing nothing.

Matrix for SWAP Gate

- The matrix of U_{SWAP} is

$$U_{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We can understand it in this way. Since this is a 2-qubit gate for operating on a vector in the four-dimensional $\mathbb{C}^4$ space, the matrix (operator) must be 4×4 . Moreover, it only swaps the second and the third basis states.

- Therefore, only the second and the third rows are different from the identity matrix. Let us try a numerical example:

$$U_{SWAP}|10\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$$

This is because $|10\rangle$ is the third basis state and thus it is $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $|01\rangle$ is $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$.

Circuit and Properties

- In general, for a 2-qubit vector, $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$, we have

$$U_{SWAP}|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha \\ \gamma \\ \beta \\ \delta \end{pmatrix}$$

$$= \alpha|00\rangle + \gamma|01\rangle + \beta|10\rangle + \delta|11\rangle$$

As expected, the coefficients of the second and the third basis states are swapped.

- Pic. 14 shows the circuit of a SWAP operation. Note that again the bottom qubit is the MSB.
- However, a quantum circuit represents a flow of time not a flow of signal. Therefore, unlike the classical swap operation, the implementation of a SWAP gate is complicated and is not as simple as how the circuit shows.
- To perform a SWAP operation, we need to apply some correct external pulses so that the states of two electron spins are swapped.



Pic.14 SWAP gate and its circuit

Phase Shift Gate

- The phase shift gate, which is also called the P-gate or the phase gate, is a 1-qubit gate that shifts the relative phase between the two basis vectors. It is defined as the following:

$$U_{PS,\phi}|0\rangle = |0\rangle$$

$$U_{PS,\phi}|1\rangle = e^{i\phi}|1\rangle$$

where ϕ is the **phase** and $e^{i\phi}$ is the **phase factor**. When the phase shift gate is applied to the basis vector $|0\rangle$, it performs nothing. But when it is applied to the basis vector $|1\rangle$, it adds an additional phase to it. Adding an additional phase means that the vector is multiplied by $e^{i\phi}$.

- Why do we only apply the additional phase to the second basis?
- What is the meaning of shifting the relative phase between the two basis vectors?
- It is easier to understand if we look at its matrix form and observe how it evolves a general vector.

Matrix for Phase Shift Gate

- The matrix of $U_{PS,\phi}$ is

$$U_{PS,\phi} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

When we apply the phase shift gate to the second basis:

$$U_{PS,\phi}|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{i\phi} \end{pmatrix} = e^{i\phi} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e^{i\phi}|1\rangle$$

Circuit and Properties

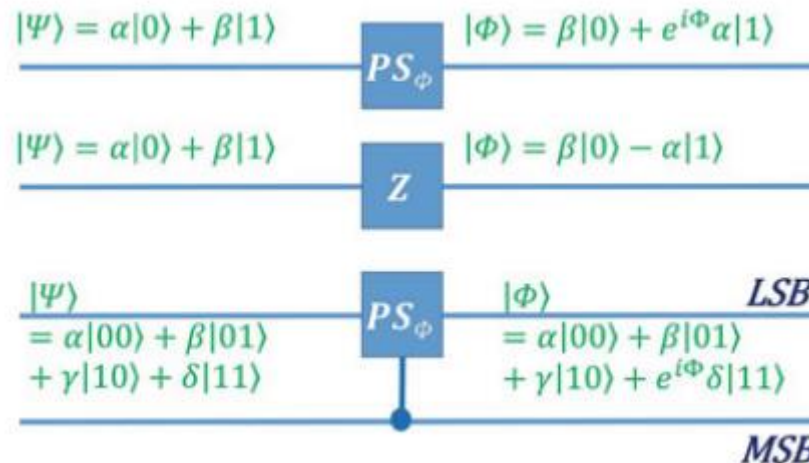
- In general, for a 1-qubit vector, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we have

$$\begin{aligned} U_{PS,\phi}|\psi\rangle &= \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ e^{i\phi}\beta \end{pmatrix} \\ &= \alpha|0\rangle + e^{i\phi}\beta|1\rangle \end{aligned}$$

- In the original vector α and β , which are complex numbers in general, can be represented as the product of a phase factor and their magnitude. For example, $\alpha = e^{i\theta_1}|\alpha|$ and $\beta = e^{i\theta_2}|\beta|$, where θ_1 and θ_2 are the phases of α and β , respectively. Therefore, their phase difference is $\theta_1 - \theta_2$.
- Due to the phase shift gate, the phase of α is unchanged but the phase of β is changed to $\phi + \theta_2$. Therefore, their relative phase is changed by ϕ . This also explains why the phase shift is only applied to the second basis state to be meaningful. If it is applied to both, there will be no change in the phase difference.
- ϕ is a parameter of the phase shift gate. When $\phi = \pi$, the gate becomes $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ because $e^{i\pi} = \cos \pi + i \sin \pi = -1$. And this is just the σ_z matrix, it is also called the **Z-gate**.
- The phase shift gate shifts the relative phase of $|0\rangle$ and $|1\rangle$ by ϕ . And thus the Z-gate shifts the relative phase by π . This makes the Z-gate turns out that in the $|+\rangle / |-\rangle$ basis. E.g. $U_{PS,\pi}|+\rangle = U_z|+\rangle = |-\rangle$.

Circuit and Properties

- We see that the Z-gate does convert $|+\rangle$ to $|-\rangle$. So, the Z-gate will add a phase shift of π to the coefficient of the second basis state (i.e. $|1\rangle$). It just multiply -1 to $|1\rangle$.
- Pic. 15 shows the circuit of a phase shift gate and Z-gate.



Pic.15 Phase shift gate circuits. Top: 1-bit phase shift gate.
 Middle: Z-gate. Bottom: controlled-PS gate

Controlled Phase Shift Gate

- Since Z-gate is the NOT gate in the $|+\rangle / |-\rangle$ basis, it is then natural that there may be a “**controlled-Z-gate**,” which is a CNOT gate in the $|+\rangle / |-\rangle$ basis. And in general, there must be a controlled phase shift gate. Indeed, we can define a 2-qubit controlled phase shift gate as the following:

$$U_{CPS,\phi}|ab\rangle = e^{i(a.b)\phi}|ab\rangle$$

where a and b are just the numbers of the first and the second qubits in the basis states, respectively. They can be either 0 or 1. Here “.” means the classical logic AND operation. Therefore, $a.b$ equals 1 only when both a and b are 1. How it changes the basis states:

$$U_{CPS,\phi}|00\rangle = e^{i(0.0)\phi}|00\rangle = |00\rangle$$

$$U_{CPS,\phi}|01\rangle = e^{i(0.1)\phi}|01\rangle = |01\rangle$$

$$U_{CPS,\phi}|10\rangle = e^{i(1.0)\phi}|10\rangle = |10\rangle$$

$$U_{CPS,\phi}|11\rangle = e^{i(1.1)\phi}|11\rangle = e^{i\phi}|11\rangle$$

- Therefore, only $|11\rangle$ is changed with an extra phase. When the first qubit is 1, a phase shift gate is applied. So, the first qubit is the control qubit and the second qubit is the target qubit in this controlled phase shift gate. the phase shift is only applied to the second basis state. Therefore, only $|11\rangle$ gets the extra phase shift.

Matrix for Controlled Phase Shift Gate

- The matrix of $U_{CPS,\phi}$ is

$$U_{CPS,\phi} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix}$$

Since this is a 2-qubit gate for operating on a vector in the four-dimensional $\mathbb{C}^4$ space, the matrix (operator) must be 4×4 . And as discussed, only $|11\rangle$ gains an extra phase after this gate. Therefore, it is almost like an identity matrix except the last row. Let us try a numerical example.

$$U_{CPS,\phi}|11\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ e^{i\phi} \end{pmatrix} = e^{i\phi} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = e^{i\phi}|11\rangle$$

Circuit and Properties

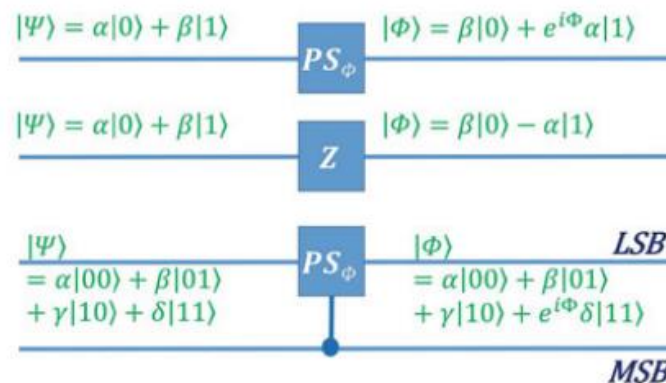
- In general, for a 2-qubit vector, $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$, we have

$$U_{CPS,\phi}|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ e^{i\phi}\delta \end{pmatrix}$$

$$= \alpha|00\rangle + \gamma|01\rangle + \beta|10\rangle + e^{i\phi}\delta|11\rangle$$

Therefore, a controlled phase shift gate only adds a phase to the $|11\rangle$ component.

- Pic. 15 shows the circuit of a controlled phase shift gate operation. Note that again the bottom qubit is the MSB.



Pic.15 Phase shift gate circuits. Top: 1-bit phase shift gate.
Middle: Z-gate. Bottom: controlled-PS gate

Toffoli (CCNOT) Gate

- A Toffoli gate is a 3-qubit gate. It is also known as the CCNOT gate, which stands for **Controlled-Controlled-NOT** gate. When a CCNOT gate is applied to the basis states of $\mathbb{C}^8$ (a 3-qubit space), it will apply a NOT operator to the third qubit if both the first and the second qubit (the control qubits) are 1. The definition is:

$$T|abc\rangle = T|a\rangle \otimes |b\rangle \otimes |c\rangle = T|a, b, c\rangle = |a, b, (a \cdot b) \oplus c\rangle$$

T is the unitary operator of the Toffoli gate. $|abc\rangle$ is the basis state of a $\mathbb{C}^8$ space, which is formed by the tensor products of three $\mathbb{C}^2$ spaces due to three qubits. a and b are either 0 or 1. $|abc\rangle$ can be any of the 8 basis states running from $|000\rangle$, $|001\rangle$, to $|111\rangle$.

- After applying T to $|a, b, c\rangle$, the basis state becomes $|a, b, (a \cdot b) \oplus c\rangle$. This means the value of the first 2 qubits remain unchanged. But the last qubit value becomes $(a \cdot b) \oplus c$. So, it is only one when both a and b are one. $\oplus$ is the classical XOR operation.
- Again $0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, and $1 \oplus 1 = 0$.

Toffoli (CCNOT) Gate

- Therefore, this is a controlled-controlled-NOT gate with the first and the second qubits being the control qubits.

$$T|0, 0, 0\rangle = |0, 0, (0.0)\oplus 0\rangle = |0, 0, 0\rangle$$

$$T|0, 0, 1\rangle = |0, 0, (0.0)\oplus 1\rangle = |0, 0, 1\rangle$$

$$T|0, 1, 0\rangle = |0, 1, (0.1)\oplus 0\rangle = |0, 1, 0\rangle$$

$$T|0, 1, 1\rangle = |0, 1, (0.1)\oplus 1\rangle = |0, 1, 1\rangle$$

$$T|1, 0, 0\rangle = |1, 0, (1.0)\oplus 0\rangle = |1, 0, 0\rangle$$

$$T|1, 0, 1\rangle = |1, 0, (1.0)\oplus 1\rangle = |1, 0, 1\rangle$$

$$T|1, 1, 0\rangle = |1, 1, (1.1)\oplus 0\rangle = |1, 1, 1\rangle$$

$$T|1, 1, 1\rangle = |1, 1, (1.1)\oplus 1\rangle = |1, 1, 0\rangle$$

We see that T keeps the basis vectors intact except swaps the last two basis vectors because only the last two basis vectors have both the first and the second qubit values being one to enable the NOT operation on the last qubit value

Matrix for Toffoli (CCNOT) Gate

- The matrix of T is

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Since the Toffoli gate is a 3-qubit gate in the eight-dimensional $\mathbb{C}^8$ space, the matrix (operator) must be 8×8 . Moreover, it only swaps the last two basis states. Therefore, it is different from an identity matrix only in the last two rows, which has the NOT gate matrix embedded. Let us try a numerical example.

$$T|111\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |110\rangle$$

Circuit and Properties

- In general, for a 3-qubit vector, $|\psi\rangle = \alpha|000\rangle + \beta|001\rangle + \gamma|010\rangle + \delta|011\rangle + \epsilon|100\rangle + \zeta|101\rangle + \eta|110\rangle + \theta|111\rangle$, we have

$$T|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \\ \epsilon \\ \zeta \\ \eta \\ \theta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \\ \epsilon \\ \zeta \\ \theta \\ \eta \end{pmatrix}$$

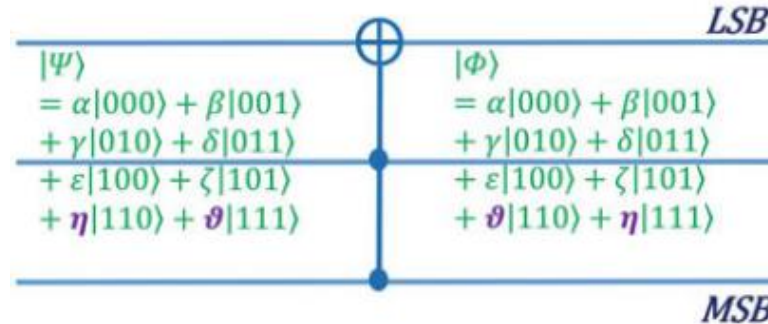
$$= \alpha|000\rangle + \beta|001\rangle + \gamma|010\rangle + \delta|011\rangle + \epsilon|100\rangle + \zeta|101\rangle + \theta|110\rangle + \eta|111\rangle$$

$$= \alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle + \delta|3\rangle + \epsilon|4\rangle + \zeta|5\rangle + \theta|6\rangle + \eta|7\rangle$$

- As expected, the coefficients of the last two basis states are swapped.

Circuit and Properties

- Pic. 16 shows the circuit of a CCNOT operation. Note that again the bottom qubit is the MSB and the two most significant qubits act as the control qubits.



Pic.16 Toffoli (CCNOT) gate circuit. The swapped coefficients are highlighted

- We can make a Toffoli gate to act as a classical AND gate by setting $c = 0$. In this case,

$$T|abc\rangle = T|ab0\rangle = |a, b, (a \cdot b) \oplus 0\rangle = |a, b, a \cdot b\rangle$$

This is because any number XOR with 0 is unchanged. Therefore, the third qubit value equals a AND b and thus it can act as a classical AND gate, with the first two qubits as the inputs and the third qubit as the output. Similarly, if we set $a = 1$ and $b = 1$, we have

$$T|abc\rangle = T|11c\rangle = |1, 1, (1 \cdot 1) \oplus c\rangle = |1, 1, 1 \oplus c\rangle = |1, 1, \bar{c}\rangle$$

This is because $1 \oplus c = \bar{c}$, where $\bar{c}$ is the negation of c .

- Since any classical logic can be implemented with a combination of the AND and NOT gates, thus, the Toffoli gate is a universal gate for classical logic.

Chapter 16:

Walsh–Hadamard Gate and Its

Properties

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Walsh–Hadamard Gate

- The **Walsh–Hadamard** gate is commonly called the **Hadamard** gate.
- The symbol of a Hadamard gate is H .
- Hadamard gate is so special because it has no classical counterpart. The Hadamard gate is a 1-qubit gate, but, of course, we can also form a multi-qubit Hadamard gate through the tensor products of 1-qubit Hadamard gates.

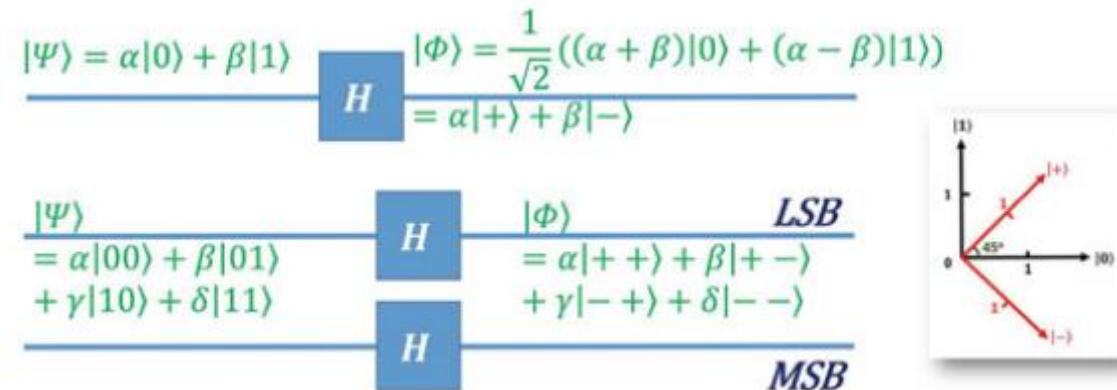
$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

- When the Hadamard gate is applied to the basis vectors $|0\rangle$ and $|1\rangle$, they are rotated to a *superposition* of the basis vectors.
- Note that when the quantum gates are applied to the basis vectors (e.g. NOT gate to $|0\rangle$), the basis vectors are only transformed (rotated) to other basis vectors (e.g. $U_{NOT}|0\rangle = |1\rangle$).
- Since the basis vectors have a one-to-one correspondence to classical logic binary values, those gates have their classical counterpart (e.g. $NOT(0) = 1$). However, there is no classical counterpart for the Hadamard gate because it is impossible to find a classical gate that gives “ $\frac{1}{\sqrt{2}}(0 + 1)$ ” as “ $0 + 1$ ” is meaningless in the classical Boolean logic.

Walsh–Hadamard Gate

- The Hadamard gate also equivalently transforms $|0\rangle$ and $|1\rangle$ to $|+\rangle$ and $|-\rangle$, respectively.
- Let's remark that $|0\rangle$ and $|1\rangle$ are the eigenvectors of σ_z and $|+\rangle$ and $|-\rangle$ are the eigenvectors of σ_x and they can be represented as the superpositions of the others.
- We used a fake real-2D plane to illustrate this idea and repeat this in the inset of Pic. 17.
- While the fake 2D plane illustration is incomplete, the equations derived turned out to be correct.
For example, using vector addition, we can easily show that $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.



Pic.17 Hadamard gate circuits. The inset shows the relationship between the $|+\rangle / |-\rangle$ basis and the $|0\rangle / |1\rangle$ basis on a fake real 2D plane.

Matrix for Walsh–Hadamard Gate

- The matrix of $\mathbf{H}$ is

$$\mathbf{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Since this is a 1-qubit gate for operating on a vector in the two-dimensional $\mathbb{C}^2$ space, the matrix (operator) must be 2×2 . And we can easily prove that this is correct by doing the numerical substitutions.

$$\begin{aligned} \mathbf{H}|0\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\ \mathbf{H}|1\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \end{aligned}$$

Because we are working on the basis formed by the eigenvectors of σ_z when we write the operator and the vectors in their matrix and column form, respectively. Therefore, $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Circuit

- In general, for a 1-qubit vector, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we have

$$\begin{aligned} \mathbf{H}|\psi\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \beta \\ \alpha - \beta \end{pmatrix} \\ &= \alpha|+\rangle + \beta|-\rangle \end{aligned}$$

This is the final expression easily by recalling how $\mathbf{H}$ rotates the basis states. That is, it will transform $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$, respectively.

Properties of the Hadamard Gate

- **Inverse of Hadamard Gate**

The inverse of the Hadamard gate (H^{-1}) equals the Hadamard gate itself, i.e.

$$H = H^{-1}$$

This is a special property that we will use in some quantum computing algorithms. Since for any matrix, when it is multiplied by its inverse, it becomes an identity matrix. Therefore, we have

$$\begin{aligned} HH^{-1} &= I \\ HH &= I \end{aligned}$$

This means that if you apply the Hadamard gate twice, the input state vector is unchanged!

The construction of the inverse of a general matrix is fair cumbersome and we will not discuss it here. But we will try to do so for a 2×2 matrix as it is relatively easy. The inverse of a 2×2 matrix, $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is

$$U^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Properties of the Hadamard Gate

- **Inverse of Hadamard Gate**

For a Hadamard gate, $a = b = c = \frac{1}{\sqrt{2}}$ and $d = -\frac{1}{\sqrt{2}}$, therefore,

$$\begin{aligned}
 \mathbf{H}^{-1} &= \frac{1}{\frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}} \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= \frac{1}{\frac{-1}{2} - \frac{1}{2}} \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \mathbf{H}
 \end{aligned}$$

Properties of the Hadamard Gate

- **Multiple-Qubit Hadamard Gate**

Multiple-qubit Hadamard gate is just a simple tensor product of 1-qubit Hadamard gates. Therefore, for an n -qubit Hadamard gate, denoted by $H^{\otimes n}$, it is found by:

$$H^{\otimes n} = H \otimes H \cdots \otimes H$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \cdots \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

For example, when $n = 2$, a 2-qubit Hadamard gate is

$$H^{\otimes 2} = H \otimes H$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & 1 \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ 1 \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & -1 \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Properties of the Hadamard Gate

- **Multiple-Qubit Hadamard Gate**

When it is applied to $|00\rangle$, i.e. the first basis vector of the $\mathbb{C}^4$ space formed by two qubits, we get

$$\begin{aligned} H^{\otimes 2}|00\rangle &= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \end{aligned}$$

It creates an *equal* superposition of all the basis vectors in the $\mathbb{C}^4$ space like in the 1-qubit case for the $\mathbb{C}^2$ space.

We have used the fact that each 1-qubit Hadamard gate is only applied to the vector in the corresponding space. It shows that, by applying the Hadamard gate to any state, we only need to replace $|0\rangle$ by $|+\rangle$ and $|1\rangle$ by $|-\rangle$.

Properties of the Hadamard Gate

- **Properties of n-Qubit Hadamard Gate**

when an n -qubit Hadamard gate ($H^{\otimes n}$) is applied to $|0\rangle_n$, it will create an equal superposition of all basis vectors in the 2^n -dimensional space (the $\mathbb{C}^{2^n}$ space). $|0\rangle_n$ is the first basis vector in the 2^n -D space. It is formed by the tensor product of n $|0\rangle$'s in the subspaces.

$$|0\rangle_n = |00 \cdots 0\rangle = |0\rangle \otimes |0\rangle \otimes \cdots \otimes |0\rangle$$

Applying an n -qubit Hadamard gate to $|0\rangle_n$, we have

$$\begin{aligned} H^{\otimes n} |0\rangle_n &= H^{\otimes n} |0\rangle \otimes |0\rangle \otimes \cdots \otimes |0\rangle \\ &= (H \otimes H \otimes \cdots \otimes H) |0\rangle \otimes |0\rangle \otimes \cdots \otimes |0\rangle \\ &= (H|0\rangle) \otimes (H|0\rangle) \otimes \cdots \otimes (H|0\rangle) \end{aligned}$$

An n -qubit Hadamard gate can be just n copies of such pulses applying to n electrons in parallel. So,

$$\begin{aligned} H^{\otimes n} |0\rangle_n &= \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \otimes \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \\ &= \frac{1}{\sqrt{2^n}} (|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \otimes \cdots \otimes (|0\rangle + |1\rangle) \end{aligned}$$

Properties of the Hadamard Gate

- **Properties of n-Qubit Hadamard Gate**

$$\begin{aligned}
 H^{\otimes n} |0\rangle_n &= \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \\
 &= \frac{1}{\sqrt{2^n}} (|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \otimes \cdots \otimes (|0\rangle + |1\rangle) \\
 &= \frac{1}{\sqrt{2^n}} (|00 \cdots 00\rangle + |00 \cdots 01\rangle + |00 \cdots 10\rangle + |00 \cdots 11\rangle + \cdots + |11 \cdots 11\rangle) \\
 &= \frac{1}{\sqrt{2^n}} (|0\rangle + |1\rangle + |2\rangle + |3\rangle + \cdots + |2^n - 1\rangle) \\
 &= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle
 \end{aligned}$$

What happen if the $H^{\otimes n}$ is applied to an arbitrary basis vector $|y\rangle$. $|y\rangle$ can be represented in a binary (e.g. $|00101\rangle$) or decimal (e.g. $|5\rangle$) form. It really does not matter how it is represented as long as we know how many qubits are being used. So, we will first represent y in its binary form.

$$|y\rangle = |y_{n-1}y_{n-2} \cdots y_1 y_0\rangle$$

Since $y_{n-1}y_{n-2} \cdots y_1 y_0$ is a string of numbers representing the value $y_{n-1} \times 2^{n-1} + y_{n-2} \times 2^{n-2} \cdots y_0 \times 2^0$. Each H only applies to the corresponding qubit. However, the qubit may be $|0\rangle$ or $|1\rangle$. And it becomes $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, respectively, after applying the Hadamard gate.

Properties of the Hadamard Gate

- **Properties of n-Qubit Hadamard Gate**

This is equivalent to

$$H|y_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{y_i}|1\rangle)$$

because $(-1)^{y_i} = 1$ and $(-1)^{y_i} = -1$ for $y_i = 0$ and $y_i = 1$, respectively. Therefore,

$$\begin{aligned} H^{\otimes n}|y\rangle &= H^{\otimes n}|y_{n-1}y_{n-2}\cdots y_1y_0\rangle \\ &= (H||y_{n-1}\rangle) \otimes (H||y_{n-2}\rangle) \otimes \cdots \otimes (H||y_0\rangle) \\ &= \left(\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{y_{n-1}}|1\rangle)\right) \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{y_{n-2}}|1\rangle)\right) \\ &\quad \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{y_0}|1\rangle)\right) \\ &= \frac{1}{2^{n/2}}(|0\rangle + (-1)^{y_{n-1}}|1\rangle) \otimes (|0\rangle + (-1)^{y_{n-2}}|1\rangle) \otimes \cdots \otimes \\ &\quad (|0\rangle + (-1)^{y_0}|1\rangle) \end{aligned}$$

$$= \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} (-1)^{x \cdot y} |x\rangle$$

Properties of the Hadamard Gate

- **Properties of n-Qubit Hadamard Gate**

For example, calculate $H^{\otimes 2}|3\rangle$

$$\begin{aligned} H^{\otimes 2}|3\rangle &= \frac{1}{2^{2/2}} \sum_{x=0}^{2^2-1} (-1)^{x \cdot 3} |x\rangle = \frac{1}{2} \sum_{x=0}^3 (-1)^{x \cdot 3} |x\rangle \\ &= \frac{1}{2} ((-1)^{0 \cdot 3} |0\rangle + (-1)^{1 \cdot 3} |1\rangle + (-1)^{2 \cdot 3} |2\rangle + (-1)^{3 \cdot 3} |3\rangle) \end{aligned}$$

For $n = 2$, we have $0_{10} = 00_2$, $1_{10} = 01_2$, $2_{10} = 10_2$, and $3_{10} = 11_2$. Therefore,

$$0.3 = (0 \text{ AND } 1) \oplus (0 \text{ AND } 1) = 0 \oplus 0 = 0$$

$$1.3 = (0 \text{ AND } 1) \oplus (1 \text{ AND } 1) = 0 \oplus 1 = 1$$

$$2.3 = (1 \text{ AND } 1) \oplus (0 \text{ AND } 1) = 1 \oplus 0 = 1$$

$$3.3 = (1 \text{ AND } 1) \oplus (1 \text{ AND } 1) = 1 \oplus 1 = 0$$

Therefore,

$$H^{\otimes 2}|3\rangle = \frac{1}{2} (|0\rangle - |1\rangle - |2\rangle + |3\rangle)$$

Chapter 17: Two Quantum Circuit Examples

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Quantum Circuit for Rotating Basis

- The standard $\mathbb{C}^2$ basis vectors are $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. They can be written in the column form as:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

For example, $|00\rangle$ is 100% of the zeroth basis vector, i.e. $|00\rangle$.

- The Bell states are given as follows:

$$|\phi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, |\phi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, |\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, |\psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

Quantum Circuit for Rotating Basis

- Find the matrix representation of $U_{XOR}(\mathbf{H} \otimes \mathbf{I})$.

$$U_{XOR}\mathbf{H} \otimes \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ 1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & -1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix}$$

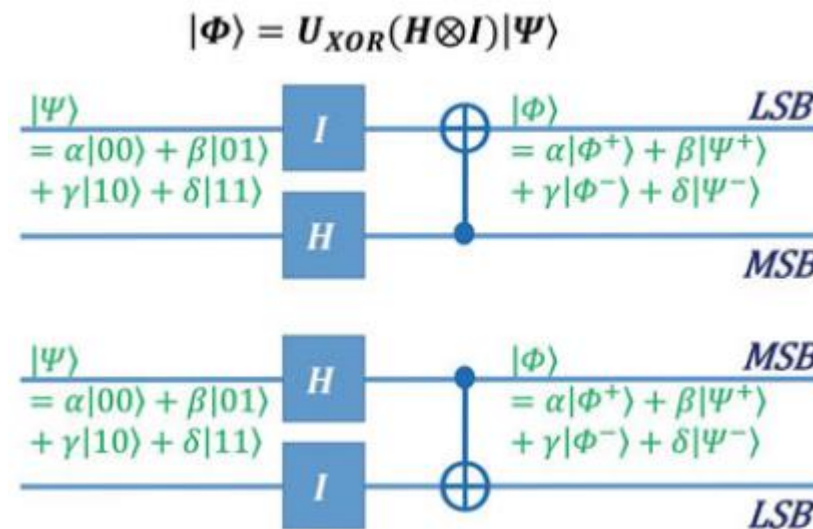
$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}$$

- Show $U_{XOR}(\mathbf{H} \otimes \mathbf{I})|01\rangle = |\psi^+\rangle$.

$$U_{XOR}\mathbf{H} \otimes \mathbf{I}|01\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = |\psi^+\rangle$$

Quantum Circuit for Rotating Basis

- How to construct a quantum circuit for this? We should note that in matrix operations, the states evolve from the right to the left in the equation, but in a quantum circuit, the states evolve from the left to the right in the circuit diagram.
- Therefore, we should put the $H \otimes I$ gate on the left and the U_{XOR} on the right in the circuit diagram.
- Pic. 18 shows the corresponding quantum circuit. The state evolves from left to right in the circuit.



Pic.18 The quantum circuits for rotating regular $\mathbb{C}^4$ basis vectors to Bell states. The top one is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books. Both are using little-endian convention

Quantum Circuit for Rotating Basis

- Here shows that $\mathbf{U}_{XOR}(\mathbf{H} \otimes \mathbf{I})(\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle) = \alpha|\phi^+\rangle + \beta|\psi^+\rangle + \gamma|\phi^-\rangle + \delta|\psi^-\rangle$.
- For the left-hand side, we have

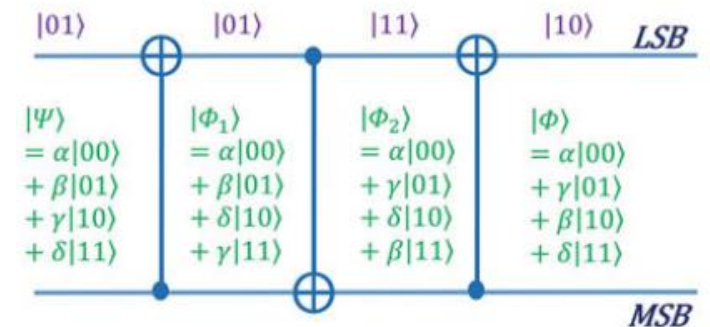
$$\begin{aligned} & \mathbf{U}_{XOR}(\mathbf{H} \otimes \mathbf{I})(\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \gamma \\ \beta + \delta \\ \beta - \delta \\ \alpha - \gamma \end{pmatrix} \end{aligned}$$

- For the right-hand side, we have

$$\begin{aligned} & \alpha|\phi^+\rangle + \beta|\psi^+\rangle + \gamma|\phi^-\rangle + \delta|\psi^-\rangle \\ &= \frac{\alpha}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \frac{\beta}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \frac{\gamma}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} + \frac{\delta}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \gamma \\ \beta + \delta \\ \beta - \delta \\ \alpha - \gamma \end{pmatrix} \end{aligned}$$

Quantum Circuit for Implementing a SWAP Gate

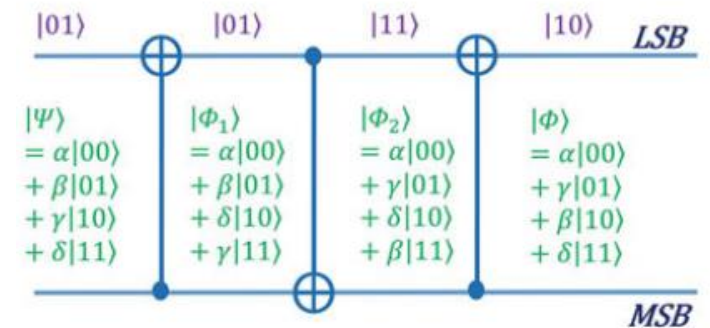
- The definition of SWAP gate is $U_{SWAP}|ab\rangle = |ba\rangle$.
- And one might think it is as simple as just crossing over two wires in a classical circuit. So maybe it is as easy as just swapping two electrons physically. We want electron 1 to acquire the state of electron 2 and electron 2 to acquire the state of electron 1 after swapping. Sometimes we may get confused and argue that the electrons are indistinguishable.
- Assume the situation that the electrons are in two different atoms or in two different quantum, which is usually the case in a real quantum computer. In this case, they are distinguishable.
- How do we swap the states of the electrons in two different atoms? This is not easy technically, especially if it is a larger system with many other electrons involved. Therefore, we need to implement a quantum circuit to perform the SWAP operation.
- Pic. 19 shows the implementation of a SWAP gate using CNOT gates.



Pic.19 Implementation of a SWAP gate using CNOT gates. The purple ket's on top shows an example of how a single basis vector evolves in the circuit

Quantum Circuit for Implementing a SWAP Gate

- The circuit in Pic.19 is composed of 3 CNOT(U_{XOR}) gates.
- The first one on the left has the bottom (MSB) as the control qubit and the top (LSB) as the target qubit. This means, for the basis vectors, if the control qubit is 0, it will do nothing to them. If the control qubit is 1, it will flip the LSB. Therefore, $|00\rangle$ and $|01\rangle$ will not be changed, but $|10\rangle$ will become $|11\rangle$ and $|11\rangle$ will become $|10\rangle$. This is equivalent to swapping the coefficients of $|10\rangle$ and $|11\rangle$ in a general vector.
- Therefore, for a general input, $|\Psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle$, and after the first CNOT gate, it becomes $|\phi_1\rangle = \alpha |00\rangle + \beta |01\rangle + \delta |10\rangle + \gamma |11\rangle$ as δ and γ are swapped.
- For the second CNOT gate, the control qubit is LSB. Therefore, it means $|00\rangle$ and $|10\rangle$ will not be changed because the LSB is 0. $|01\rangle$ will become $|11\rangle$ and $|11\rangle$ will become $|01\rangle$ because the LSB is 1 and it will flip the MSB. Therefore, for a general vector, it will swap the coefficients of $|01\rangle$ and $|11\rangle$. So the β and γ in $|\phi_1\rangle$ are swapped and it becomes $|\phi_2\rangle = \alpha |00\rangle + \gamma |01\rangle + \delta |10\rangle + \beta |11\rangle$.
- Finally, the rightmost CNOT gate swaps the coefficients of $|10\rangle$ and $|11\rangle$ just as the leftmost CNOT gate. So the β and δ in $|\phi_2\rangle$ are swapped and it becomes $|\phi\rangle = \alpha |00\rangle + \gamma |01\rangle + \beta |10\rangle + \delta |11\rangle$.



Pic.19 Implementation of a SWAP gate using CNOT gates. The purple ket's on top shows an example of how a single basis vector evolves in the circuit

Quantum Circuit for Implementing a SWAP Gate

- We know that a CNOT gate swaps the coefficients of $|10\rangle$ and $|11\rangle$ if the MSB is the control qubit and it swaps the coefficients of $|01\rangle$ and $|11\rangle$ if the LSB is the control qubit. Both of them involve $|11\rangle$. Therefore, we can treat $|11\rangle$ as a “middleman” or a “buffer” to complete the swapping using only CNOT gates.
- For example, $U_{XOR, right} U_{XOR, center} U_{XOR, left}$, where “right,” “center,” and “left” indicate the locations of the CNOT gate in the circuit in Pic. 19. The matrix representation of the circuit is:

$$\begin{aligned}
 U_{XOR, right} U_{XOR, center} U_{XOR, left} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = U_{SWAP}
 \end{aligned}$$

Again, in an equation, the input vector evolves from the right to the left. Therefore, the matrix of the leftmost CNOT in the circuit appears at the rightmost position of the equation.